🌐 Global Internet & Technology Adoption Analysis

📊 This project explores global trends in digital technology adoption using datasets from the World Bank.

We analyze how the following indicators have evolved across countries:

🌐 Internet users (% population)
📱 Mobile cellular subscriptions
📡 Fixed broadband subscriptions
💰 GDP per capita
👥 Population

🎯 The goal is to understand the **global digital divide** and the relationship between **technology access and economic development**.

🎯 Project Objectives

The main objectives of this analysis are:

🔎 Understand global trends in internet and digital technology adoption.

📊 Explore relationships between economic development and digital connectivity.

🌍 Identify countries with the highest levels of technological adoption.

📈 Investigate whether economic strength correlates with digital infrastructure.

💡 Provide insights into the global digital divide.

📥 Data Source

All datasets used in this project were obtained from the **World Bank Open Data portal**.
https://databank.worldbank.org/source/world-development-indicators#

The datasets provide country-level indicators collected from national statistical systems and international organizations.

The indicators used in this analysis include:

- Internet users (% of population)
- Mobile cellular subscriptions
- Fixed broadband subscriptions
- GDP per capita

These indicators allow us to analyze both **technological adoption** and **economic performance** across countries and over time.

📂 Dataset Structure

Each dataset initially followed the **World Bank standard format**, where years were represented as individual columns.

Example structure:

Country | Country Code | Indicator Name | 1990 | 1991 | ... | 2025

While this format is useful for storage, it is not ideal for analysis and visualization.

Therefore, the datasets were later transformed into a more analytical structure.

In [ ]:

In [1]:
```
import numpy as np;
import pandas as pd;
import seaborn as sns;
import matplotlib.pyplot as plt;
import plotly.express as px;

sns.set_style("whitegrid");
```

🧹 Phase 1 — Data Cleaning

The raw datasets required preprocessing before analysis.

The following cleaning steps were performed:

- ✔ Removed unnecessary metadata columns such as indicator codes and names.
- ✔ Eliminated columns containing entirely missing values.
- ✔ Handled missing values using statistical imputation methods.

Numeric columns were filled using the **median**, while categorical columns were filled using the **mode**.

This approach ensures that missing data does not significantly distort the statistical properties of the dataset.

In [2]:
```
Internet=pd.read_csv("Part-1(py).csv",engine='python',encoding='latin1',skiprows=4);
Cellular=pd.read_csv("Part-3(py).csv",engine='python',encoding='latin1',skiprows=4);
Subscribed=pd.read_csv("Part-4(py).csv",engine='python',encoding='latin1',skiprows=4);
GDP=pd.read_csv("Part-5(py).csv",engine='python',encoding='latin1',skiprows=4);
```

In [3]:
```
def cleaning_data(df):
    df=df.dropna(axis=1,how="all");
    df=df.drop(columns=["Indicator Code","Indicator Name"],errors="ignore");
    return df;
```

In [4]:
```
Internet=cleaning_data(Internet);
Cellular=cleaning_data(Cellular);
Subscribed=cleaning_data(Subscribed);
GDP=cleaning_data(GDP);
```

In [5]:
```
Internet.columns
```

Out[5]:
```
Index(['Country Name', 'Country Code', '1990', '1991', '1992', '1993', '1994',
       '1995', '1996', '1997', '1998', '1999', '2000', '2001', '2002', '2003',
       '2004', '2005', '2006', '2007', '2008', '2009', '2010', '2011', '2012',
       '2013', '2014', '2015', '2016', '2017', '2018', '2019', '2020', '2021',
       '2022', '2023', '2024', '2025'],
      dtype='object')
```

🔄 Phase 2 — Data Transformation

The datasets were originally stored in **wide format**, where each year appeared as a separate column.

Example:

Country | 1990 | 1991 | 1992 | ... | 2025

To enable efficient analysis, the data was transformed into **long format** using a reshaping operation.

New structure:

Country | Year | Indicator Value

This transformation allows easier merging of datasets and supports time-series analysis.

In [6]:
```
def convert_long(df,value_name):
    year_cols = df.filter(regex=r'^\d{4}$').columns;

    df_long = df.melt(
        id_vars=["Country Name","Country Code"],
        var_name="Year",
        value_vars=year_cols,
        value_name=value_name
    )
    df_long["Year"]=df_long["Year"].astype(int);
    return df_long;
```

In [7]:
```
Internet_long = convert_long(Internet, "Internet")
Cellular_long = convert_long(Cellular, "Cellular")
```

```
Subscribed_long = convert_long(Subscribed, "Broadband")
GDP_long = convert_long(GDP, "GDP")
```

In [8]:
```
print(Internet_long.head(10));
```

```
                 Country Name Country Code  Year  Internet
0                       Aruba          ABW  1990       0.0
1  Africa Eastern and Southern          AFE  1990       NaN
2                 Afghanistan          AFG  1990       0.0
3   Africa Western and Central          AFW  1990       NaN
4                      Angola          AGO  1990       0.0
5                     Albania          ALB  1990       0.0
6                     Andorra          AND  1990       0.0
7                  Arab World          ARB  1990       NaN
8         United Arab Emirates          ARE  1990       0.0
9                   Argentina          ARG  1990       0.0
```

🔗 Phase 3 — Dataset Integration

After transformation, each indicator dataset was merged into a single master dataset.

The merge operation was performed using the following common keys:

- Country Name
- Country Code
- Year

This integration produced a unified dataset containing multiple indicators for each country-year combination.

The final dataset allows comparative analysis across economic and technological variables.

In [9]:
```
df = Internet_long.merge(Cellular_long, on=["Country Name","Country Code","Year"]);
df = df.merge(Subscribed_long, on=["Country Name","Country Code","Year"]);
df = df.merge(GDP_long, on=["Country Name","Country Code","Year"]);
```

In [10]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7182 entries, 0 to 7181
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Country Name  7182 non-null   object
 1   Country Code  7182 non-null   object
 2   Year          7182 non-null   int64
 3   Internet      5795 non-null   float64
 4   Cellular      6469 non-null   float64
 5   Broadband     5143 non-null   float64
 6   GDP           6933 non-null   float64
dtypes: float64(4), int64(1), object(2)
memory usage: 392.9+ KB
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
Country Name       0
Country Code       0
Year               0
Internet        1387
Cellular         713
Broadband       2039
GDP              249
dtype: int64
```

🧹 Phase 4 — Advanced Data Cleaning

Additional preprocessing steps were applied to ensure data quality.

These included:

- ✔ Converting indicator columns to numeric formats.
- ✔ Identifying potential outliers using statistical methods.
- ✔ Flagging extreme values without removing them to preserve meaningful economic differences.

Outlier detection helps highlight countries that significantly deviate from global patterns, such as economies with exceptionally high GDP levels.

In [12]:

```
def cleaning (df):
    '''handling nulls'''
    null_cols=df.columns[df.isnull().any()];
    for col in null_cols:
        if df[col].dtype in ["int64","float64"]:
            df[col].fillna(df[col].median(),inplace=True);
        elif df[col].dtype in ["object","str"]:
            df[col].fillna(df[col].mode()[0],inplace=True);

    '''handling outliers'''
    num_cols=df.select_dtypes(include=np.number).columns;
    for col in num_cols:
        skew_val=df[col].skew();
        if(abs(skew_val>1)):
            # Choosing IQR Method...
            Q1=df[col].quantile(0.25);
            Q3=df[col].quantile(0.75);
            IQR=Q3-Q1;

            #Ranging...
            lower=Q1-1.5*IQR;
            upper=Q3+1.5*IQR;
            #Flagging...
            df[col+"_outlier"]=((df[col]<lower) | (df[col]>upper)).astype(int);
            #Capping...
            df[col]=np.where(df[col]<lower,lower,df[col]);
            df[col]=np.where(df[col]>upper,upper,df[col]);
        else:
            #Z-Score Method...
            mean_val=df[col].mean();
            std_val=df[col].std();
            score=(df[col]-mean_val)/std_val;

            #Flagging...
            df[col+"_outlier"]=(abs(score)>3).astype(int);
            #Capping...
            df[col]=np.where(score>3,mean_val+3*std_val,df[col]);
            df[col]=np.where(score<-3,mean_val-3*std_val,df[col]);

    '''handling Skewness'''
    for col in num_cols:
```

```
        skew_val=df[col].skew();
        #For right skewed data...
        if((skew_val)>1 and (df[col]>0).all()):
            df[col]=np.log1p(df[col]);
        #For left skewed data...
        elif(skew_val)<-1:
            df[col]=np.sqrt(df[col].max()-df[col]);

    return df

cleaning(df);
```

In [13]:
```
print(df.isnull().sum());
```

```
Country Name         0
Country Code         0
Year                 0
Internet             0
Cellular             0
Broadband            0
GDP                  0
Year_outlier         0
Internet_outlier     0
Cellular_outlier     0
Broadband_outlier    0
GDP_outlier          0
dtype: int64
```

In [14]:
```
print(df.columns);
```

```
Index(['Country Name', 'Country Code', 'Year', 'Internet', 'Cellular',
       'Broadband', 'GDP', 'Year_outlier', 'Internet_outlier',
       'Cellular_outlier', 'Broadband_outlier', 'GDP_outlier'],
      dtype='object')
```

📊 Dataset Ready for Analysis

After completing the cleaning, transformation, and integration steps, the dataset is now prepared for exploratory analysis.

The final dataset contains:

- Country identifiers
- Year information
- Technology adoption indicators
- Economic indicators

This structure allows us to explore global trends and relationships between technology adoption and economic development.

In [ ]:

🎯 Objective 1: Analyze the Global Growth of Internet Adoption

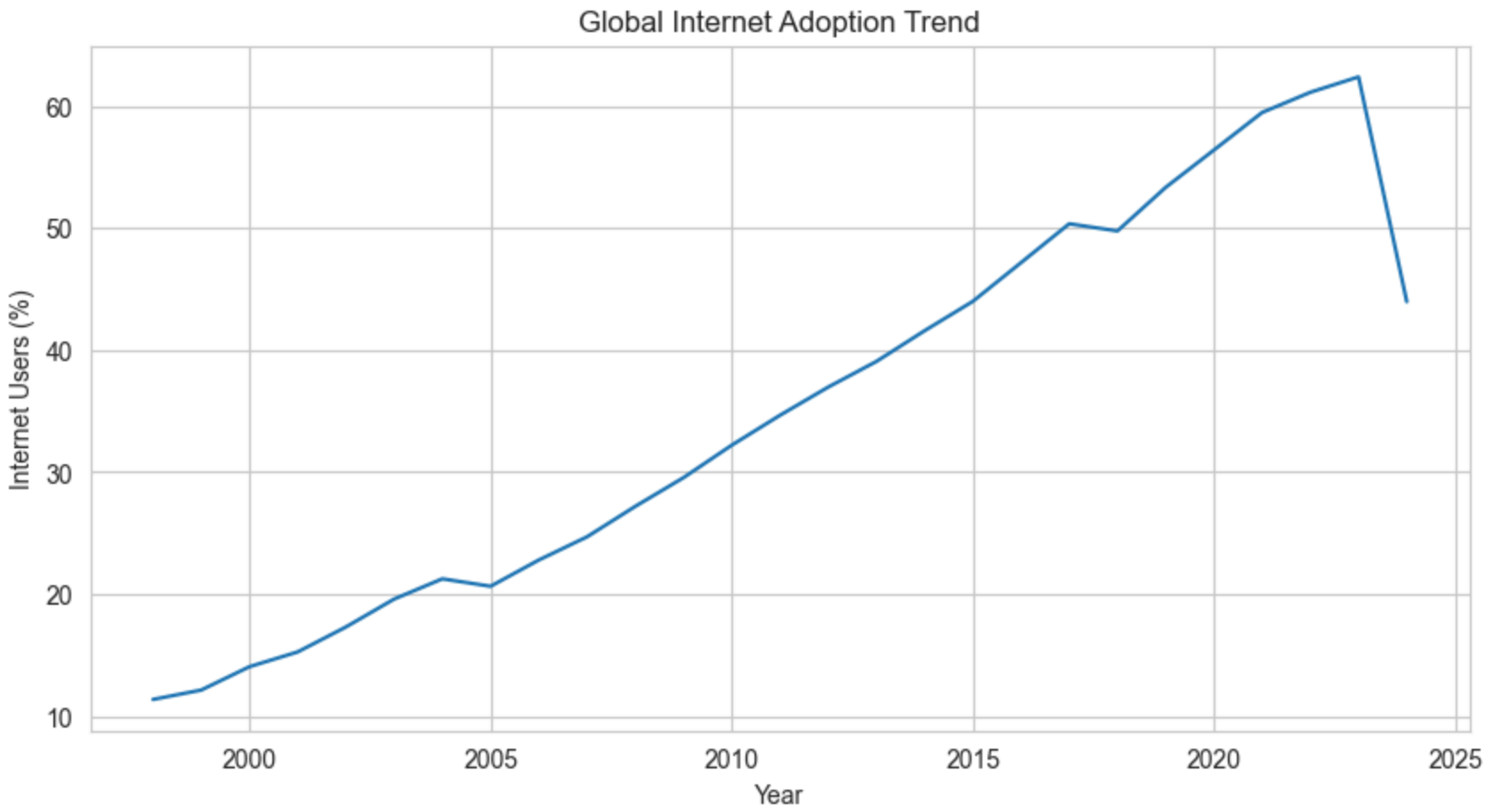
🌐 Over the past few decades, internet connectivity has expanded dramatically across the world.

In this section, we analyze the **average global internet penetration over time** to understand how quickly digital connectivity has grown.

This provides a baseline for understanding the global digital transformation.

In [15]:
```
trend = df.groupby("Year")["Internet"].mean().reset_index();
```

In [16]:
```
plt.figure(figsize=(10,5));
sns.lineplot(data=trend, x="Year", y="Internet");
plt.title("Global Internet Adoption Trend");
plt.ylabel("Internet Users (%)");
plt.show();
```



💡 **Insight** Observe the acceleration of internet growth and identify periods of rapid digital expansion.

In [ ]:

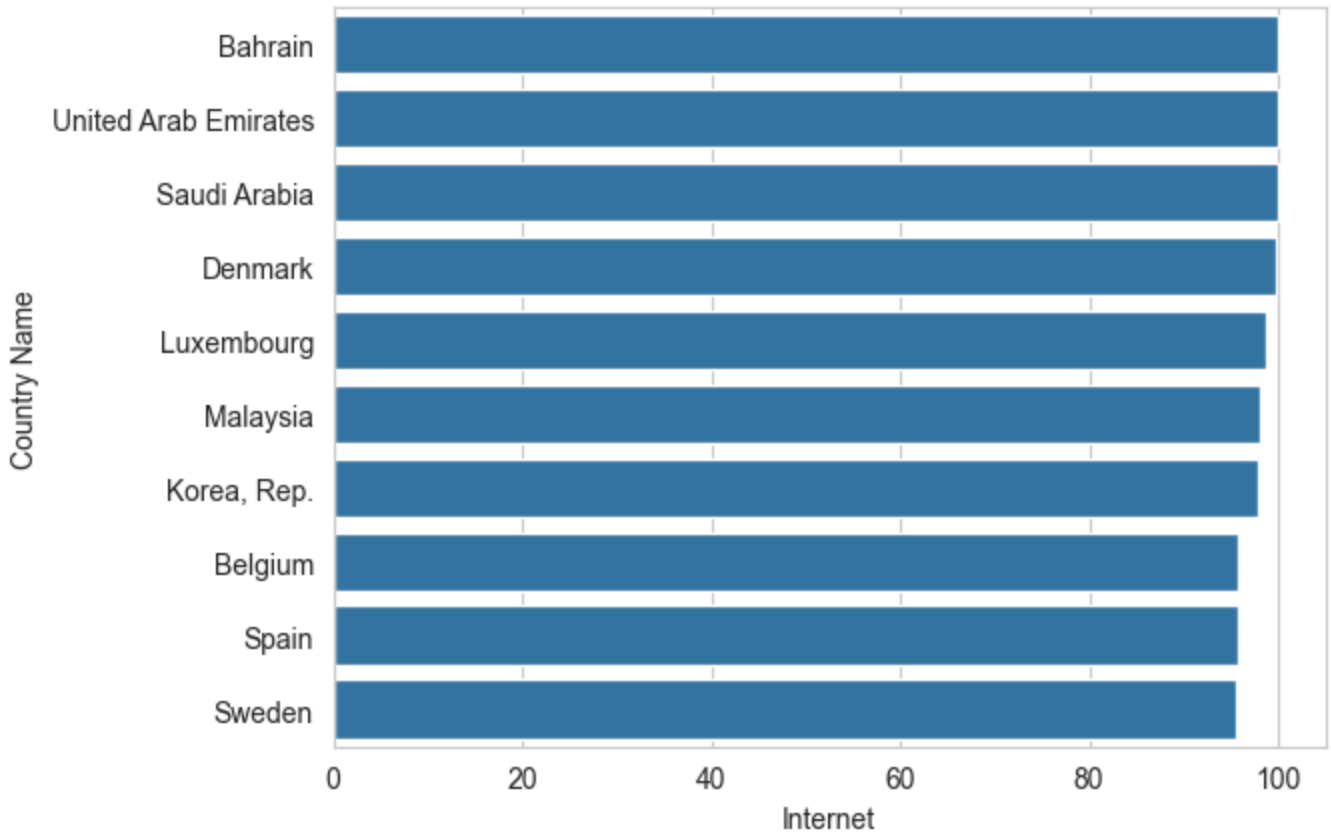
🎯 Objective 2: Identify Countries with the Highest Internet Penetration

🌍 Although global internet usage has increased significantly, adoption levels vary across countries.

This analysis highlights the **countries with the highest internet penetration**, allowing us to identify global digital leaders.

In [17]:
```
latest = df[df["Year"] == df["Year"].max()];
top_countries = latest.sort_values("Internet", ascending=False).head(10);
```

In [18]:
```
sns.barplot(data=top_countries, x="Internet", y="Country Name");
```



💡 **Insight** These countries represent the most digitally connected societies.

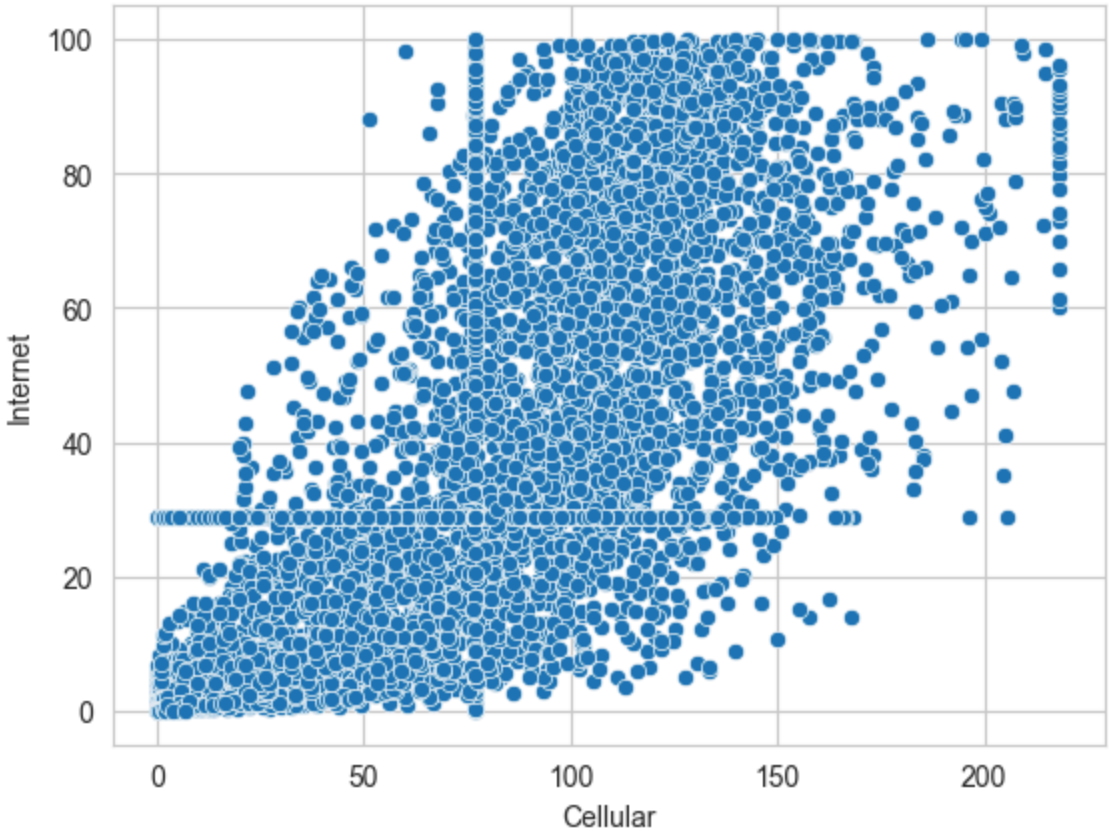
In [ ]:

🎯 Objective 3: Investigate the Role of Mobile Connectivity in Internet Adoption

📱 Mobile networks have been a key driver of internet expansion, particularly in developing economies.

This section explores whether countries with higher **mobile cellular subscriptions** also tend to have higher **internet penetration rates**.

In [19]:
```
sns.scatterplot(data=df, x="Cellular", y="Internet");
```



💡 **Insight** A strong positive relationship would suggest that mobile infrastructure significantly contributes to internet accessibility.

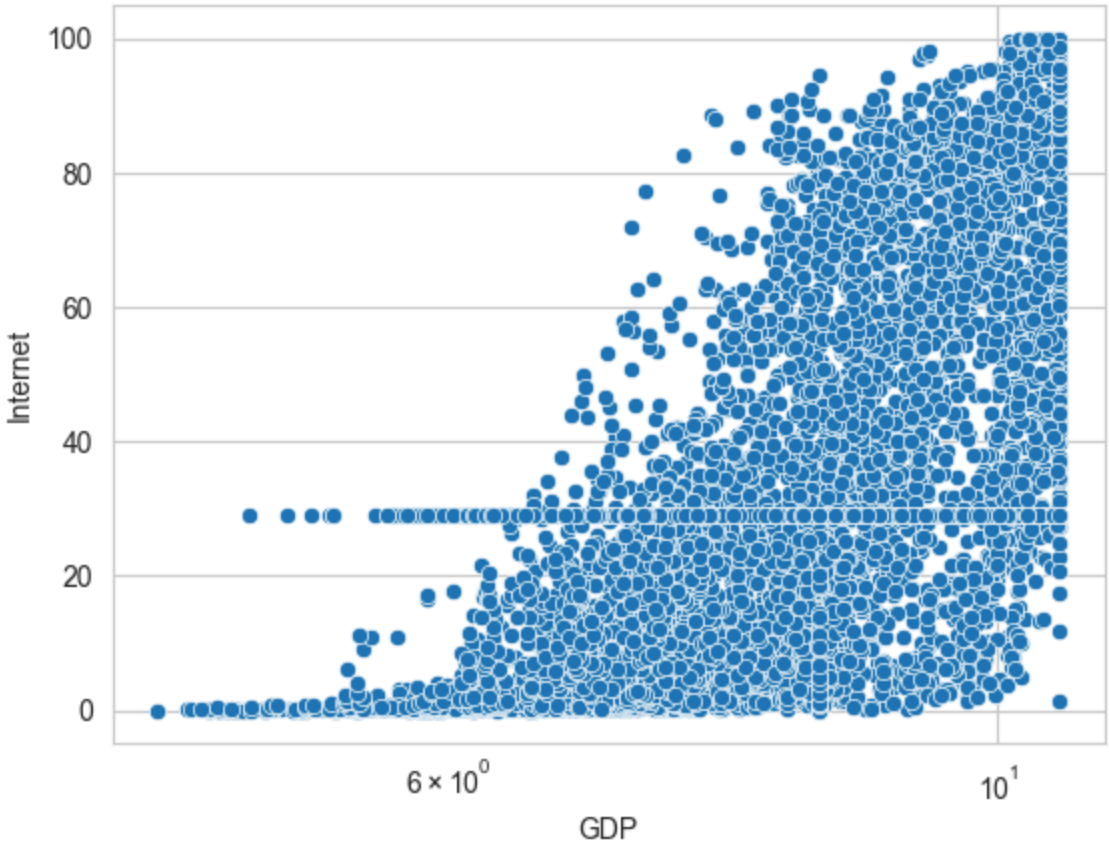
In [ ]:

🎯 Objective 4: Explore the Relationship Between GDP and Internet Usage

💰 Economic development often influences the availability of technological infrastructure.

In this analysis, we examine whether countries with higher GDP levels tend to exhibit higher internet penetration rates.

In [20]:
```
sns.scatterplot(data=df, x="GDP", y="Internet");
plt.xscale("log");
```



💡 **Insight** This visualization helps reveal whether economic prosperity contributes to greater digital connectivity.

In [ ]:

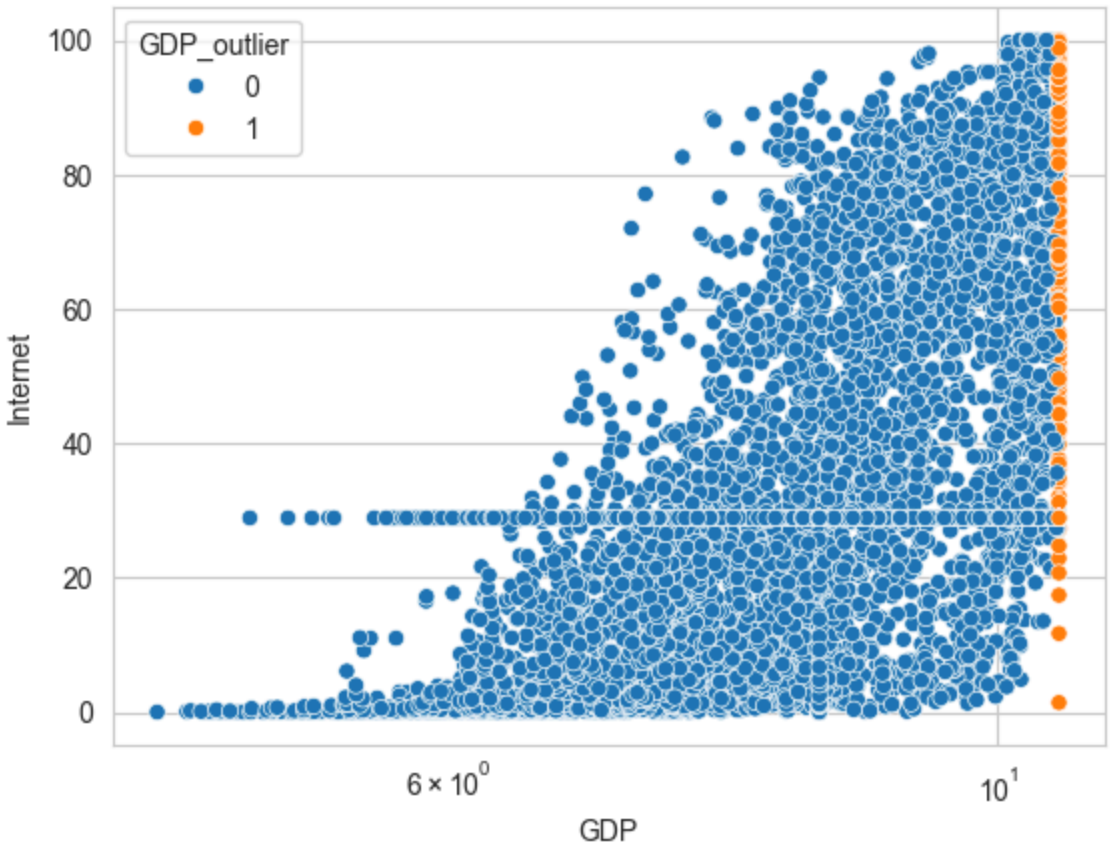
🎯 Objective 5: Identify Countries with Exceptional Economic Performance

🌍 Some countries exhibit GDP levels that significantly exceed the global norm.

Using the outlier flags generated during data preprocessing, we identify **countries with extreme GDP values** and analyze their digital connectivity characteristics.

In [21]:
```
gdp_outliers = df[df["GDP_outlier"] == 1];
```

In [22]:
```
sns.scatterplot(data=df, x="GDP", y="Internet", hue="GDP_outlier");
plt.xscale("log");
```



💡 **Insight** This helps determine whether economically dominant countries also lead in digital connectivity.

In [ ]:

🎯 Objective 6: Compare Digital Connectivity Between Big Economic and Normal Countries

💰 Economic power often influences the development of digital infrastructure.
Countries with exceptionally high GDP levels may have greater resources to invest in connectivity, telecommunications, and digital technologies.

Using the **GDP outlier flags** created during the preprocessing stage, we compare digital connectivity indicators between:

- Economically dominant countries (GDP outliers)
- The rest of the global economy

This comparison helps determine whether economic strength translates into stronger digital infrastructure.

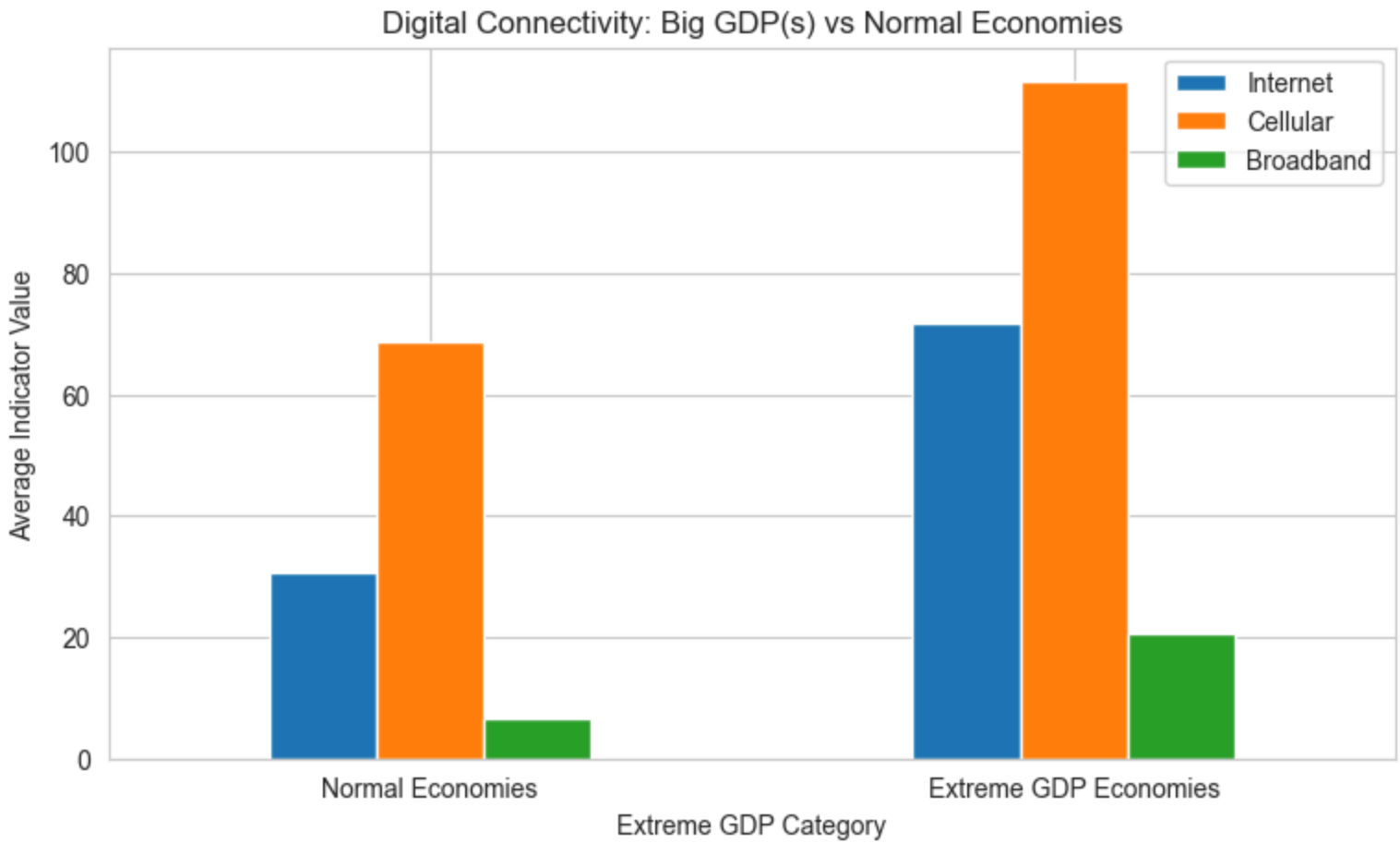
In [23]:
```
comparison = df.groupby("GDP_outlier")[["Internet","Cellular","Broadband"]].mean().reset_index();

comparison;
```

In [24]:
```
comparison.set_index("GDP_outlier").plot(
    kind="bar",
    figsize=(9,5)
);

plt.title("Digital Connectivity: Big GDP(s) vs Normal Economies")
plt.ylabel("Average Indicator Value");
plt.xlabel("Extreme GDP Category");
plt.xticks([0,1],["Normal Economies","Extreme GDP Economies"], rotation=0);

plt.show();
```



💡 **Insight**

The comparison highlights clear differences between economically dominant countries and the rest of the world.

Countries identified as GDP outliers generally exhibit higher levels of internet penetration, mobile connectivity, and broadband infrastructure.

This suggests that economic strength plays an important role in accelerating digital adoption and infrastructure development.

In [ ]:

🌐 Internet Usage Prediction using Machine Learning

In this section, we build a machine learning model to predict **Internet usage (% of population)** based on key economic and technological indicators.

🎯 Objective

To understand how factors like:

- GDP (economic growth)
- Cellular subscriptions
- Broadband subscriptions

influence the adoption of internet across countries.

📊 Features & Target

- **Features (Independent Variables):**
  - GDP
  - Cellular Subscriptions
  - Broadband Subscriptions
- **Target (Dependent Variable):**
  - Internet Usage

⚙️ Approach

- Data cleaning and preprocessing
- Train-test split (80-20)
- Feature scaling (for better model performance)
- Model training using:
  - Linear Regression

    - Random Forest Regressor
- Performance evaluation using:
    - R^2 Score
    - Mean Absolute Error (MAE)

🧠 Goal

To build a predictive model and analyze which factors most strongly drive internet adoption.

In [25]:
```
df
```

Out[25]:

	Country Name	Country Code	Year	Internet	Cellular	Broadband	GDP	Year_outlier	Internet_outlier	Cellular_outlier	Bro
0	Aruba	ABW	1998.0	28.9000	6.377808	4.38436	9.843158	0	0	0	
1	Africa Eastern and Southern	AFE	1998.0	28.9000	0.963135	4.38436	6.548333	0	0	0	
2	Afghanistan	AFG	1998.0	28.9000	0.000000	4.38436	8.474251	0	0	0	
3	Africa Western and Central	AFW	1998.0	28.9000	0.098026	4.38436	7.040962	0	0	0	
4	Angola	AGO	1998.0	0.0185	0.063904	4.38436	6.064223	0	0	0	
...	...	...	...	...	...	...	...	...	...	...	
7177	Kosovo	XKX	2024.0	28.9000	76.883600	4.38436	8.857098	0	0	0	
7178	Yemen, Rep.	YEM	2024.0	28.9000	76.883600	4.38436	8.474251	0	0	0	
7179	South Africa	ZAF	2024.0	28.9000	76.883600	4.38436	8.743242	0	0	0	
7180	Zambia	ZMB	2024.0	28.9000	76.883600	4.38436	7.080119	0	0	0	
7181	Zimbabwe	ZWE	2024.0	28.9000	76.883600	4.38436	7.823327	0	0	0	

7182 rows × 12 columns

In [26]:

```
df['Internet_prev'] = df.groupby('Country Name')['Internet'].shift(1);
df = pd.get_dummies(df, columns=['Country Name'], drop_first=True);

features = ["GDP", "Cellular", "Broadband", "Internet_prev"]+[col for col in df.columns if col.startswith("Country_"
target = "Internet";

ml_df = df[features + [target]].dropna();
X = ml_df[features];
y = ml_df[target];
```

```python
#Train-Test Split...
from sklearn.model_selection import train_test_split;
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42);

from sklearn.preprocessing import StandardScaler;
scaler = StandardScaler();
X_train = scaler.fit_transform(X_train);
X_test = scaler.transform(X_test);

#Training...
from sklearn.ensemble import RandomForestRegressor;
model = RandomForestRegressor(n_estimators=300, max_depth=None, random_state=42);
model.fit(X_train, y_train);

#Prediction...
y_pred = model.predict(X_test)
#Evaluation...
from sklearn.metrics import r2_score, mean_absolute_error;
print("Final R2 Score:", r2_score(y_test, y_pred));
print("Final MAE:", mean_absolute_error(y_test, y_pred));

#Feature Importance...
importance = model.feature_importances_;
feat_imp = pd.DataFrame({'Feature': features,'Importance': importance}).sort_values(by='Importance', ascending=False)
print("\nFeature Importance:\n", feat_imp);
```

```
Final R2 Score: 0.9331027520649746
Final MAE: 3.0729723556235897

Feature Importance:
          Feature  Importance
3  Internet_prev    0.943244
1       Cellular    0.026986
0            GDP    0.020384
2      Broadband    0.009386
```

🚀 Final Insights

- The model achieved a high R^2 score of 0.93, indicating strong predictive performance.
- The lag feature (previous year's internet usage) dominates the model, contributing over 94% importance.

- This highlights a strong temporal dependency in internet adoption trends.
- Economic and infrastructure factors (GDP, Cellular, Broadband) play a secondary but meaningful role.

⚠️ Key Observation

The model relies heavily on historical data, suggesting that internet growth follows a momentum-based pattern rather than being solely driven by external factors.

In [ ]:

```

```

✅ Conclusion

This analysis explored global patterns of digital technology adoption using key indicators such as **internet usage, mobile cellular subscriptions, broadband connectivity, and GDP**, combined with a machine learning approach to predict internet growth.

Several important insights emerged from the study:

📈 **Strong Temporal Growth Pattern**
Internet adoption follows a highly time-dependent trend. The machine learning model revealed that **previous internet usage (lag feature)** is the most significant predictor of future adoption, indicating a strong momentum effect in digital growth.

🌍 **Uneven Digital Development**
Despite global progress, internet penetration remains uneven across countries, highlighting the presence of a persistent **global digital divide** influenced by regional and structural differences.

📱 **Mobile Connectivity as a Key Driver**
Mobile subscriptions show a meaningful relationship with internet usage, suggesting that mobile infrastructure plays a critical role in expanding digital access, especially in developing regions.

💰 **Economic Influence on Digital Infrastructure**
Countries with higher GDP levels generally demonstrate stronger digital connectivity, reinforcing the idea that economic strength supports technological advancement.

🧠 **Model Insight: Momentum over External Factors**
Feature importance analysis showed that while GDP and telecom infrastructure contribute to predictions, **historical internet usage dominates**, indicating that digital adoption is largely driven by its own past growth rather than only external variables.

🚩 **Economic Outliers and Digital Leadership**
Countries with strong economic performance tend to lead in digital adoption, but the model highlights that **consistent growth over time** is equally important as economic power.

📊 Final Takeaway

The study demonstrates that **internet adoption is not only influenced by economic and infrastructural factors but is also strongly driven by temporal momentum**.

While investments in GDP and telecom infrastructure are essential, sustained growth patterns play a crucial role in shaping future digital expansion.

Bridging the digital divide will require not only economic investment and policy support but also long-term, consistent development strategies to ensure inclusive digital transformation across all regions.

In [ ]:

📌 *Data Source: World Bank Open Data (World Development Indicators)*

(1) https://data.worldbank.org/indicator/IT.NET.USER.ZS

(2) https://data.worldbank.org/indicator/NY.GDP.PCAP.CD

(3) https://data.worldbank.org/indicator/IT.NET.BBND.P2

(4) https://data.worldbank.org/indicator/IT.CEL.SETS.P2

(5) https://data.worldbank.org/indicator/IT.NET.USER.ZS

In [ ]: